

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE						
Regular/Supplementary Summer Examination – 2025						
Course: B.Tech		Branch : AI & DS / Allied		Semester :VII		
Subject Code & Name: Open Elective-Block Chain Technology (BTAIOE705B)						
Max Marks: 60		Date:25-7-2025		Duration: 3 Hr.		
Instructions to the Students: 1. Each question carries 12 marks. 2. Question No. 1 will be compulsory and include objective-type questions. 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6. 4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 5. Use of non-programmable scientific calculators is allowed. 6. Assume suitable data wherever necessary and mention it clearly.						
					(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)					12
1	What does P2P stand for?				CO1	1
	a. Peer to Public	b. Public to Public	c. Public to Peer	d. Peer to Peer		
2	How does blockchain prevent double-spending?				CO1	1
	a. Two-factor authentication	b. Encryption alone	c. Consensus mechanisms (e.g., PoW/PoS)	d. Central authority verification		
3	What is “gas” in Ethereum?				CO4	1
	a. A type of decentralized app	b. A token standard like ERC-20	c. A fee paid in ETH to execute transactions/s mart contracts	d. The unit of account in Ethereum		
4	What is a key benefit of using blockchain for land registry?				CO5	1
	a. Centralized data control	b. Offline recordkeepin g only	c. Immutable, tamper-proof records	d. Slower transfer times		
5	How many wei are there in 1 ether?				CO4	1
	a. 10^9	b. 10^12	c. 10^15	d. 10^18		
6	What flaw in cross-border supply chain payments can blockchain address?				CO4	1

	a. Manual invoicing	b. High costs & slow settlement	c. Excessive automation	d. Lack of security		
7	What is a “message digest”?				CO2	1
	a. Encrypted message content	b. Fixed-size hash of a message	c. Digital signature	d. Key exchange protocol		
8	Which of the following is part of a block's header?				CO3	1
	a. Previous block hash	b. Merkle Root	c. Nonce	d. All of the above		
9	Which is <i>not</i> a consensus mechanism?				CO3	1
	a. Practical Byzantine Fault Tolerance (PBFT)	b. Proof of Work (PoW)	c. Proof of Spending (PoSpen)	d. Proof of Stake (PoS)		
10	Which type of blockchain is open to anyone, enabling full transparency?				CO1	1
	a. Private	b. Consortium	c. Public	d. Hybrid		
11	Which feature ensures blocks cannot be changed once added?				CO1	1
	a. Consensus	b. Smart Contracts	c. Hashing	d. Merkle trees		
12	Which of the following is NOT a key feature of blockchain?				CO1	1
	a. Decentralization	b. Centralization	c. Immutability	d. Transparency		
Q. 2	Solve the following.					12
A)	Explain the fundamental features of blockchain technology that distinguish it from traditional databases.				CO1	6
B)	What is Double Spending explain with suitable example.				CO1	6
Q.3	Solve the following.					12
A)	What is Hash Function? Explain its characteristics.				CO2	6
B)	Explain Merkle Tree and Merkle Tree Structure.				CO2	6
Q. 4	Solve Any Two of the following.					12

A)	What is Bitcoin? What are the advantages and disadvantages of Bitcoin as Currency?	CO3	6
B)	Write a difference between PoW and PoS.	CO3	6
C)	Explain Genesis Block in Block chain technology.	CO3	6
Q.5	Solve Any Two of the following.		12
A)	Explain Dapps.	CO4	6
B)	What is Ethereum? Give it's comparison with Bitcoin.	CO4	6
C)	Explain EVM.	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	Investigate the role of blockchain in ensuring Food Security. How can it be applied to trace the origin and quality of food products, enhancing overall food supply chain management?	CO5	6
B)	Explain Blockchain enabled Cross Border Payments.	CO5	6
C)	Analyze the potential benefits and challenges of implementing a Voting System on Blockchain. How can it enhance transparency and security in electoral processes?	CO5	6
	*** End ***		